

Hospital Analytics SQL Project

Database Design and Business Analytics using MySQL



Project Purpose

Hospitals generate extensive operational and clinical data. This project applies practical SQL techniques to organize that data, run analytics, and surface actionable insights for administrators and care teams.



Audience

Designed for recruiters, technical interviewers, data analysts, professors, and GitHub reviewers evaluating SQL ability and database design for healthcare contexts.



Outcome

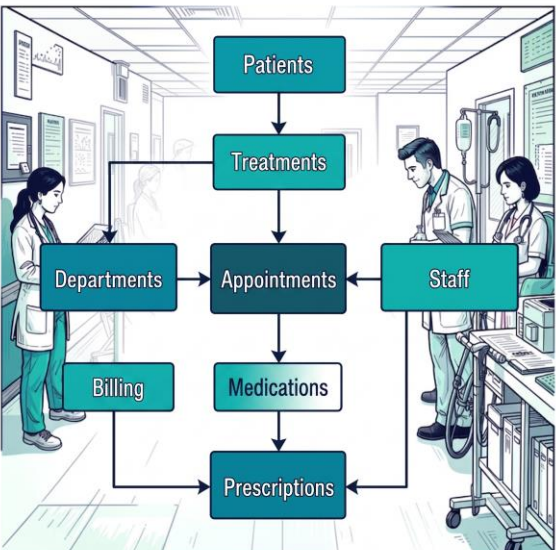
A production-style relational model, 30+ business queries, diagnostic analytics, revenue reporting, and interview-ready SQL examples demonstrating advanced window functions and query optimization.

Database Schema (Overview)

Relational model with 8 normalized tables. Primary keys (PK) and foreign keys (FK) enforce relationships, support referential integrity, and enable joins across patient care, clinical events, and billing.

- Patients (PK: patient_id)
- Doctors (PK: doctor_id)
- Departments (PK: dept_id)
- Appointments (PK: appointment_id; FK: patient_id, doctor_id)
- Treatments (PK: treatment_id; FK: appointment_id)
- Billing (PK: bill_id; FK: appointment_id)
- Medicines (PK: medicine_id)
- Prescriptions (PK: prescription_id; FK: patient_id, medicine_id, doctor_id)

Design notes: Indexed PKs for join performance; FK constraints for data integrity; lookup tables for departments and payment methods.



Technologies

MySQL, MySQL Workbench, CSV datasets, CLI and GUI tools for schema management and query execution.



Key SQL Concepts

Joins, Aggregate Functions, GROUP BY, HAVING, Subqueries, Window Functions, Indexing, and Query Optimization.



Deliverables

Normalized schema, ER diagram, 30+ documented SQL queries, sample CSV data, and analytical reports (revenue, utilization, top diagnoses).

SQL Concepts Implemented

SELECT / WHERE / ORDER BY

Row selection, filtering, and deterministic ordering for reports and exploration.

GROUP BY / HAVING / Aggregates

Summaries by department, doctor, and payment method using COUNT, SUM, AVG.

INNER JOIN / Multi-table Joins

Combining patients, appointments, treatments, and billing records for end-to-end analytics.

Subqueries & CTEs

Pre-aggregation and modular query composition for clarity and performance.

Window Functions

RANK(), DENSE_RANK(), SUM() OVER(), ROW_NUMBER() to compute running totals and top-n partitions.

Business Problems Solved

- Count total patients and active registrations (time-bounded).
- Identify busiest doctors by appointment volume and billings.
- Department-wise doctor counts and utilization rates.
- Hospital revenue analysis by date, department, and payment method.
- Most prescribed medicines and prescription frequency.
- Most common diagnoses and seasonal trends.
- List patients with multiple appointments or repeat visits.
- Top-3 doctors per department using window functions for performance reviews.



Advanced SQL Features (Interview Focus)



RANK() / DENSE_RANK()

Identify ties and ordered partitions (e.g., top doctors by revenue).



SUM() OVER()

Compute running revenue totals for trend analysis and cumulative KPIs.



Top-N Per Group

Retrieve top-3 doctors per department using window functions; efficient and interview-relevant.

Project Outcomes & KPIs

Relational Database Built

8 normalized tables with referential integrity and indexes.

30+ Business Queries

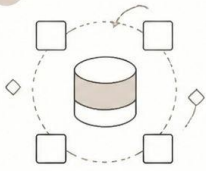
Documented queries covering operational, clinical, and financial analytics.

Meaningful Insights

Revenue by department, busiest clinicians, medication utilization, and repeat patient trends.

Key Learnings

01



Database Design

02

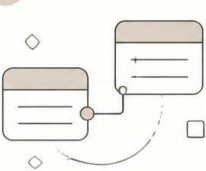
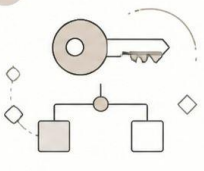


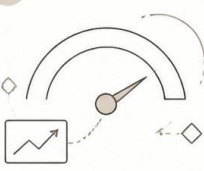
Table Relationships

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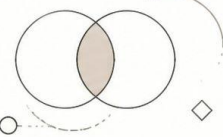
Primary & Foreign Keys

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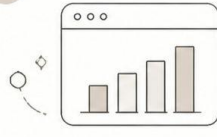
Query Optimization

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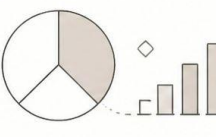
Joins

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Window Functions

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Business Analytics Thinking

These learning outcomes improve candidate readiness for SQL-focused interviews and practical analytics tasks in healthcare settings.

Thank You

Prepared by: Meet Chovatiya — Hospital Analytics SQL Project. Repository includes schema DDL, sample CSVs, MySQL Workbench model, and a query notebook with explanations and execution notes.

